

# Day 19 - Lecture 18.5: Multiple Datasets on Line Plot

---

## 1. Overview & Purpose

---

A single line plot shows trend, but comparing two or more series on the same axes enables **comparative performance analysis**:

- How do Oscar Winners compare to Non-Oscar Blockbusters in revenue?
  - How does Product A compare to Product B?
- 

## 2. Core Concepts & Parameters

---

To plot multiple lines on the same axes:

1. Call `plt.plot()` sequentially for each dataset.
2. Supply a descriptive `label` to each call:

```
plt.plot(x, y1, label="Oscar Winners")  
plt.plot(x, y2, label="Non-Oscar Winners")
```

3. Call `plt.legend()` to display the key mapping colors to labels.

### Legend Placement ( `loc` parameter):

- `'best'` (default, Matplotlib automatically chooses least overlapping spot)
  - `'upper right'`, `'upper left'`, `'lower right'`, `'lower left'`
  - `'center'`, `'center left'`, `'center right'`
-

### 3. Code Implementation

---

```
import matplotlib.pyplot as plt

years = [2008, 2009, 2010, 2011, 2012]
oscar_revenue = [1005, 170, 427, 133, 232]      # Oscar Winners
non_oscar_revenue = [378, 2788, 829, 185, 275]  # Non-Oscar Blockbusters (Avatar)

plt.figure(figsize=(9, 5))
plt.plot(years, oscar_revenue, label="Oscar Winners", marker='o', linewidth=2, color='blue')
plt.plot(years, non_oscar_revenue, label="Non-Oscar Winners", marker='s', linewidth=2, color='red')

plt.title("Revenue Comparison: Oscar vs Non-Oscar Winners", fontsize=14, fontweight='bold')
plt.xlabel("Years", fontsize=12)
plt.ylabel("Revenue (in $M)", fontsize=12)
plt.legend(loc="upper right", fontsize=11)
plt.grid(True, linestyle=":", alpha=0.6)
plt.show()
```

---

### 4. Key Takeaways

- Without `plt.legend()`, labels defined in `plt.plot(..., label="...")` will not appear on the chart.
- Ensure both datasets share the same x-axis values or are plotted over compatible ranges.